

Exercises with Time Series

EES 4891/5891

Probability & Statistics for Geosciences

Jonathan Gilligan

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Learning Goals

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- Learn about using unit-root tests to determine whether a time-series is stationary or nonstationary.
- Learn how to use the Box-Jenkins method to identify for $AR(p)$, $MA(q)$, and $ARMA(p, q)$ processes
 - Estimate p and q for $AR(p)$ and $MA(q)$ processes
- Learn how to use R functions from the *feasts* and *modeltime* packages to fit parameters for $AR(p)$, $MA(q)$, and $ARMA(p, q)$ models and to identify the best model to describe a process.

Getting Started

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- Accept the GitHub Classroom assignment at <https://classroom.github.com/a/90GLihzx>



AR, MA, and ARMA

Definitions

- $AR(p)$

$$(y_i - \mu) = \sum_{j=1}^p \phi_j (y_{i-j} - \mu) + \varepsilon_i$$

- $MA(q)$

$$y_i - \mu = \sum_{k=1}^q \theta_k \varepsilon_{i-k} + \varepsilon_i$$

- $ARMA(p, q)$

$$(y_i - \mu) = \sum_{j=1}^p \phi_j (y_{i-j} - \mu) + \sum_{k=1}^q \theta_k \varepsilon_{i-k} + \varepsilon_i$$

Tests for Stationarity

- We use “unit-root” tests for stationarity of $AR(p)$ processes:
 - The reference is to the *characteristic equation* for the autoregressive process with coefficients ϕ_i
 - The characteristic equation is a polynomial of degree p
 - For a stationary process, all the roots of the characteristic equation will be less than one.
- All $MA(q)$ processes are stationary, as long as $\varepsilon \sim \mathcal{N}(0, \sigma)$
- Unit-root tests in R:
 - For a data frame df with columns t for the time and y for the value.
 - Use the package `feasts`
 - Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test: `unitroot_kpss(df$y)`
 - Null hypothesis H_0 : stationary
 - Alternate hypothesis H_a : nonstationary
 - Phillips-Perron (PP) test: `unitroot_pp(df$y)`
 - Null hypothesis H_0 : nonstationary
 - Alternate hypothesis H_a : stationary
 - This can be confusing because we have to interpret the tests in opposite ways:
 - KPSS: $p < 0.05$ means nonstationary
 - PP: $p < 0.05$ means stationary

Box-Jenkins Method

Box-Jenkins Method

- Plot ACF and PACF

- Stationarity:

- Stationary: ACF decays to zero fairly quickly
- Nonstationary: ACF decays very slowly

- AR(p):

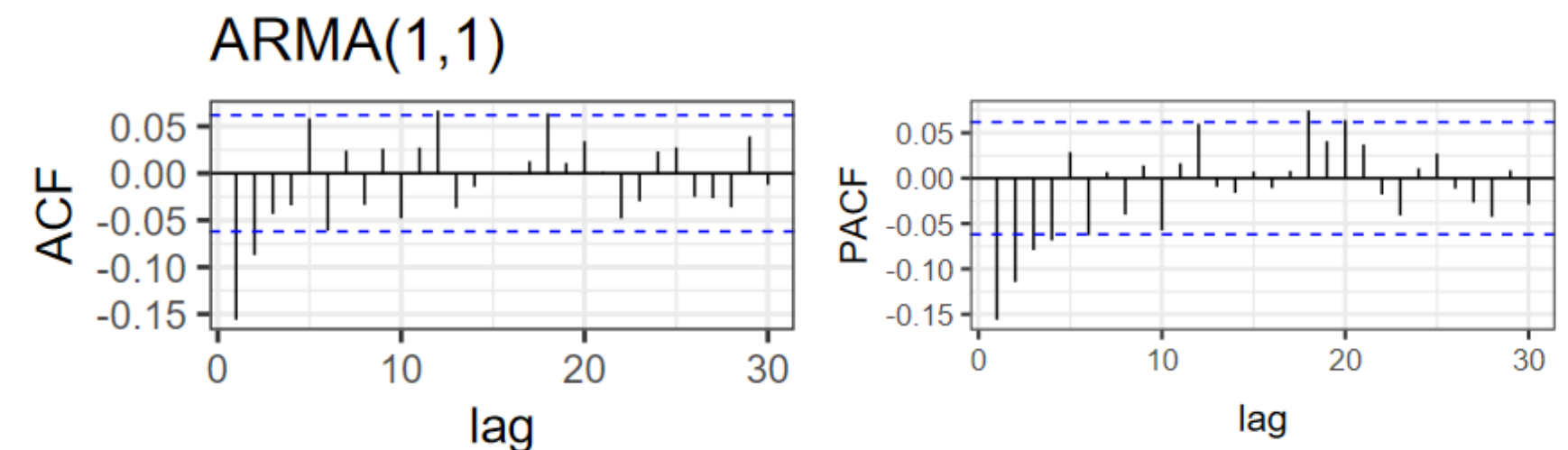
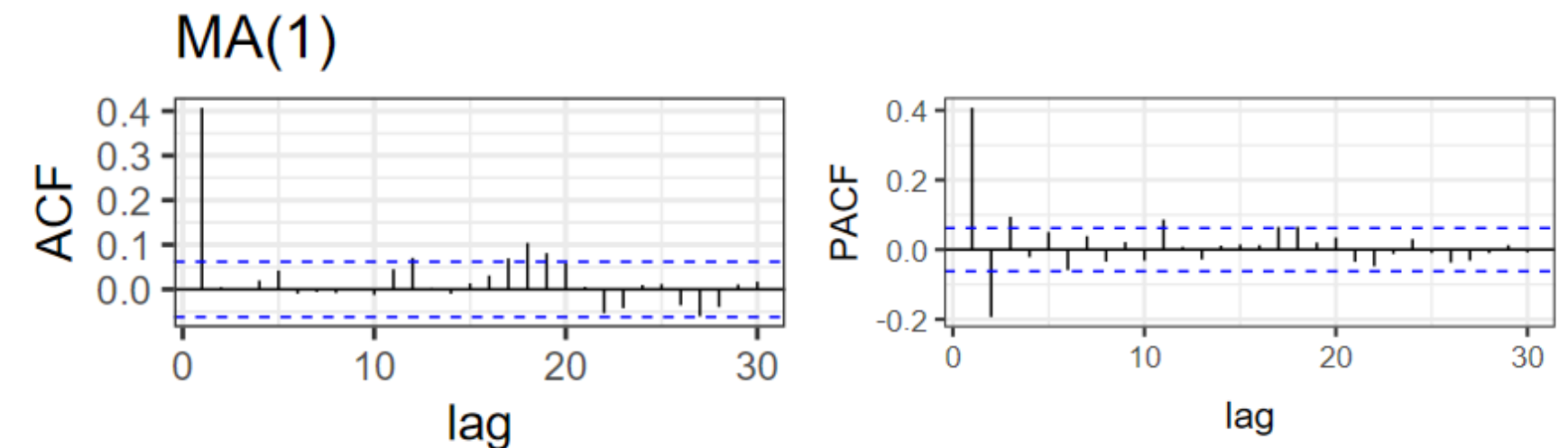
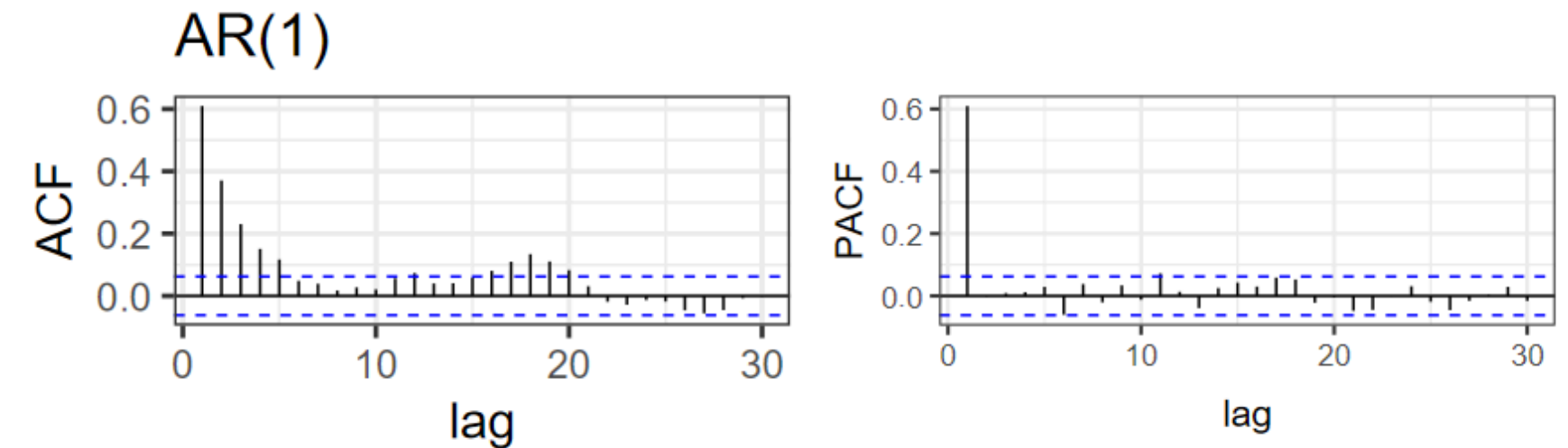
- ACF is decaying exponential or oscillation
- PACF is significant at lag p , maybe earlier too, but nothing after p

- MA(q):

- PACF is decaying exponential or oscillation
- ACF is significant at lag q , maybe earlier too, but nothing after q

- ARMA(p,q):

- Both ACF and PACF are decaying exponential or oscillation



ARMA Processes

- $AR(p)$ and $MA(q)$ processes let you estimate p or q from ACF and PACF using Box-Jenkins
- It's hard to estimate p and q for an ARMA process
 - Use regression and estimate AIC for different p and q .
 - Choose the model with smallest AIC
 - Use cross-validation and estimate R^2 and RMSE,
 - Choose the model with largest R^2 or smallest RMSE.

- R: `fable` and `feasts` packages:

```
library(fable)
library(feasts)
```

- Automatically select model:

```
model(df_arma, ARIMA(y ~ pdq(0:5, 0, 0:5)))
```

```
## # A mable: 1 x 1
##   `ARIMA(y ~ pdq(0:5, 0, 0:5))`
##                                     <model>
## 1                                <ARIMA(1,0,1) w/ mean>
```

- Compare models:

```
model(df_arma, arma_11 = ARIMA(y ~ pdq(1, 0, 1)),
      arma_21 = ARIMA(y ~ pdq(2, 0, 1)),
      arma_12 = ARIMA(y ~ pdq(1, 0, 2)),
      arma_22 = ARIMA(y ~ pdq(2, 0, 2))) |>
  glance() |> arrange(AIC)
```

```
## # A tibble: 4 x 8
##   .model  sigma2 log_lik   AIC   AICc   BIC ar_roots  ma_roots
##   <chr>    <dbl>  <dbl>  <dbl>  <dbl>  <dbl> <list>    <list>
## 1 arma_11 0.00249  1578. -3149. -3149. -3129. <cpl [1]> <cpl [1]>
## 2 arma_21 0.00249  1579. -3148. -3148. -3124. <cpl [2]> <cpl [1]>
## 3 arma_12 0.00250  1578. -3148. -3148. -3128. <cpl [1]> <cpl [2]>
## 4 arma_22 0.00250  1578. -3146. -3146. -3121. <cpl [2]> <cpl [2]>
```

feasts vs. modeltime

- There are two important families of packages for statistical modeling of time series:
 - `tidyverts` by Rob Hyndeman, consists of `tsibble`, `fable`, and `feasts`. These packages provide one way to integrate time-series models into the `tidyverse`.
 - They are very powerful, but don't integrate with the `tidymodels` framework.
 - To use the model-analysis in `fable` and `feasts`, you need to convert your data frames into `tsibbles`, which extend the `tibble` data frame into a time-series
- `modeltime` integrates time-series models with `tidymodels`, but it isn't as flexible about time-series:
 - All time series have to convert the time variable to `date` or `datetime` formats.
 - They don't work with time-series that have numerical values for the time variable.
 - They have powerful functions for cross-validation and model selection that are not available in the `tidyverts` framework.
 - The `timetk` package has a lot of useful functions for working with `modeltime` time-series.

